VELO OVERSPEED

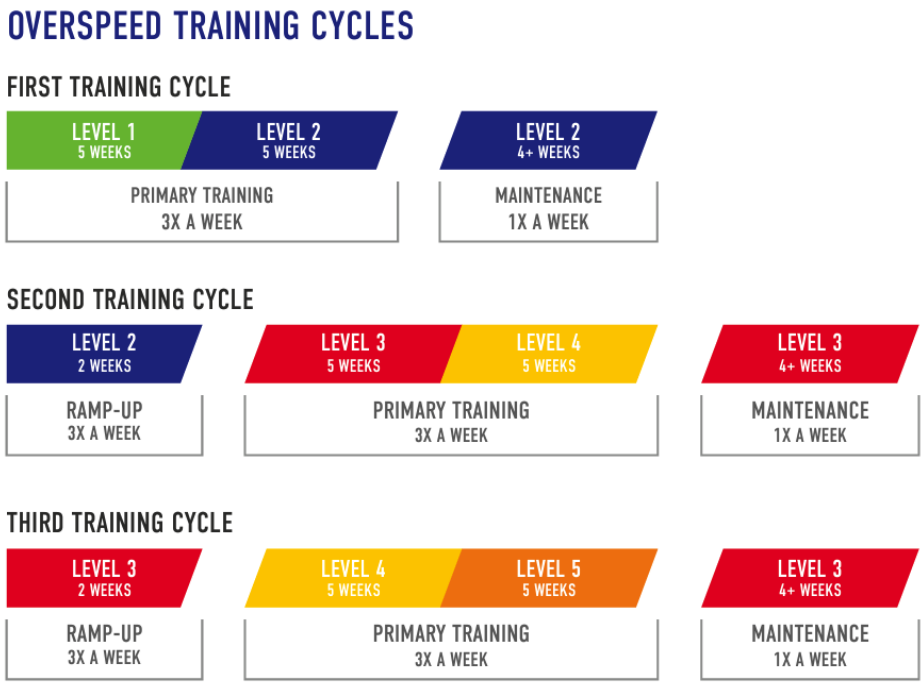
OverSpeed Training and Protocol Overview

Our goal is to help players increase bat speed. The process to accomplish this goal centers around neurological speed training using our primary OverSpeed Bat System. Extensive research has shown that there is an ideal range of bat weights relative to the player's actual game bat needed to optimize this training. The average of this range is:

Bat 1 - 20% Lighter than the game bat
Bat 2 - 10% Lighter than the game bat
Bat 3 - 5% Heavier than the game bat

Most traditional Overload/Underload Training in baseball has used bats significantly heavier than our system. For pure speed training purposes, bats heavier than 10% above the game bat have been found to commonly make the player's bat speed slow down during the training, therefore making these bats much less effective.

At Velo Sports, we have five levels of OverSpeed Training Protocols, organized into three Training Cycles. OverSpeed Training does not produce linear results. We typically see a jump in speed after the very first session of training. It then takes about 6-8 weeks for this increase in speed to become permanent for the player. Our Primary Training Phases are 10 weeks long to ensure that each player reaches this point of permanency with their gains.



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We recommend that players train 3X per week with a full day off in between sessions during primary training phases. More than this will cause the player to see lower in session speeds and get a diminished overall result from the training.

We also incorporate Maintenance Phases after the Primary Training Phase. These periods are a time for the athlete to reduce the frequency of the training from 3X per week down to 1X per week. We have found these periods should last at least 4 weeks, but can be extended to as much as a full season without any degradation of the results.

The Second and Third Training Cycles begin with a Ramp Up Phase. This is simply a two week period where the player will re-acclimate to 3X a week training out of a maintenance phase, before shifting to more advanced protocols.

OverSpeed Training Protocols

Notes:

- We always recommend that players go through our dynamic warm-up before training.
- Players will always take a 60-90 second break before the HEAVY set during each drill.
- The player should always be trying to create maximum bat speed.

OverSpeed Level 1

Drill	Sets	Reps	Side	Desc
Normal Stance	3	3	Both	Starting with just the base bat, make 3 swings on the dominant side, then 3 swings on the non-dominant side. Repeat with the First Sleeve for Set 2 Repeat with Both Sleeves for Set 3
Step Through Swings	3	3	Both	With just the base bat, make 3 swings on the dominant side lifting the lead foot and then stepping through the swing aggressively. Repeat this on the non-dominant side. Repeat with the First Sleeve for Set 2 Repeat with Both Sleeves for Set 3
Max Out	1	3	Dom	With your normal motion, make 3 swings with just the base bat. Go for your personal bests!

VELO OVERSPEED

OverSpeed Level 2

Drill	Sets	Reps	Side	Desc
Normal Stance	3	3	Both	Starting with just the base bat, make 3 swings on the dominant side, then 3 swings on the non-dominant side. Repeat with the First Sleeve for Set 2 Repeat with Both Sleeves for Set 3
Step Back Swings	3	3	Both	Starting with your feet together and just the base bat, make a small counter-rotation forward, then step and plant the back foot, then make your swing. Repeat this motion on the non-dominant side. Repeat with the First Sleeve for Set 2 Repeat with Both Sleeves for Set 3
Step Through Swings	3	3	Both	With just the base bat, make 3 swings on the dominant side lifting the lead foot and then stepping through the swing aggressively. Repeat this on the non-dominant side. Repeat with the First Sleeve for Set 2 Repeat with Both Sleeves for Set 3
Max Out	1	3	Dom	With your normal motion, make 3 swings with just the base bat. Go for your personal bests!

VELO OVERSPEED

OverSpeed Level 3

Drill	Sets	Reps	Side	Desc
Normal Stance	3	3	Both	Starting with just the base bat, make 3 swings on the dominant side, then 3 swings on the non-dominant side. Repeat with the First Sleeve for Set 2 Repeat with Both Sleeves for Set 3
Rocker Swings	3	3	Both	Starting with just the base bat, lift the front foot, then back foot and repeat until you feel you can create a maximum speed swing. Repeat this process on the non-dominant side. Repeat with the First Sleeve for Set 2 Repeat with Both Sleeves for Set 3
Double Step	3	3	Both	Starting with your feet together and just the base bat, allow a small counter-rotation, step back with the trail foot and plant, then let the lead foot come up and plant before the swing. Repeat this process on the non-dominant side. Repeat with the First Sleeve for Set 2 Repeat with Both Sleeves for Set 3
Max Out	1	3	Dom	With your normal motion, make 3 swings with just the base bat. Go for your personal bests!

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OverSpeed Level 4

Drill	Sets	Reps	Side	Desc
Normal Stance	3	3	Both	Starting with just the base bat, make 3 swings on the dominant side, then 3 swings on the non-dominant side. Repeat with the First Sleeve for Set 2 Repeat with Both Sleeves for Set 3
Double Step	3	3	Both	Starting with your feet together and just the base bat, allow a small counter-rotation, step back with the trail foot and plant, then let the lead foot come up and plant before the swing. Repeat this process on the non-dominant side. Repeat with the First Sleeve for Set 2 Repeat with Both Sleeves for Set 3
Sprint	3	5	Dom	Starting with the base bat only, make 5 swings in fast succession without ever letting the bat stop moving. Repeat with the First Sleeve for Set 2 Repeat with Both Sleeves for Set 3
Max Out	1	3	Dom	With your normal motion, make 3 swings with just the base bat. Go for your personal bests!

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OverSpeed Level 5

Drill	Sets	Reps	Side	Desc
Normal Stance	3	3	Both	Starting with just the base bat, make 3 swings on the dominant side, then 3 swings on the non-dominant side. Repeat with the First Sleeve for Set 2 Repeat with Both Sleeves for Set 3
Wind Up	3	3	Both	Starting with just the base bat, make a very aggressive movement with the lead foot allowing the knee to kick up and plant aggressively during the swing. Repeat on the non-dominant side. Repeat with the First Sleeve for Set 2 Repeat with Both Sleeves for Set 3
Crow-Hop	3	5	Dom	Starting with the base bat only, hold the bat in front of you while you make a crow-hop motion and bring the bat back. Once you plant firmly on the lead side, swing aggressively through. Repeat with the First Sleeve for Set 2 Repeat with Both Sleeves for Set 3
Max Out	1	3	Dom	With your normal motion, make 3 swings with just the base bat. Go for your personal bests!

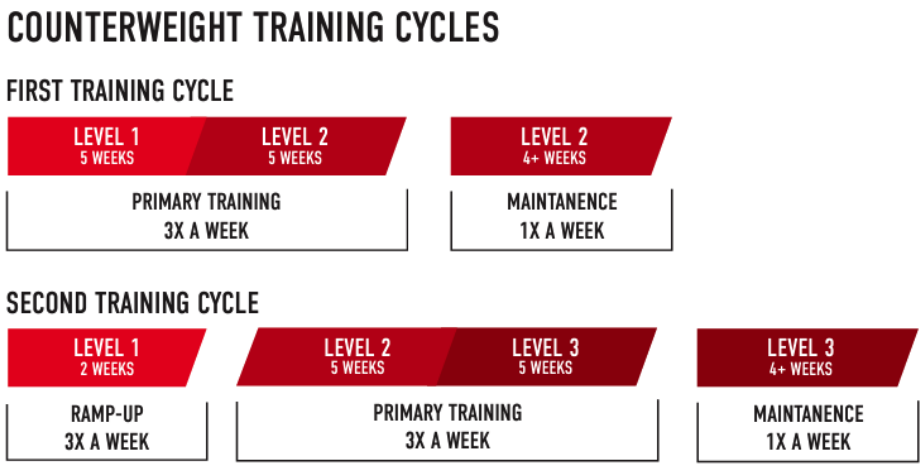
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Counterweight Training Protocols

Our Velo Counterweight Training Protocols will use the Velo OverSpeed Bat System in conjunction with the Velo Puck. The goal here will be to create a significantly Handle Weighted bat for the player to use while working on hand speed and bat delivery mechanics.

Counterweighting the bat will reduce the MOI and shift the balance point toward the player's hands. This not only allows for more bat speed, but also changes the distribution of mass of the bat significantly. These training protocols are designed to help players keep the bat level during the zone as long as possible as well as increasing bat speed.

Counterweight Training also works in Cycles like our OverSpeed Training Protocols.



Notes:

- We always recommend that players go through our dynamic warm-up before training.
- Players will always take a 60-90 second break before the HEAVY set during each drill.
- The player should always be trying to create maximum bat speed.
- For each corresponding Velo OverSpeed Bat set, we recommend adding the Velo Puck to the system along with the First Weight Sleeve.

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Counterweight Level 1

Drill	Sets	Reps	Side	Desc
Normal Stance	3	3	Both	Starting with just the base bat, make 3 swings on the dominant side, then 3 swings on the non-dominant side. Repeat with the First Sleeve for Set 2 Repeat with Both Sleeves for Set 3
Long Zone Swings	3	3	Both	With just the base bat, make 3 swings on the dominant side trying to keep the bat as flat as possible during the entire motion. Try having the knob of the bat slightly higher to start and keep the Velo Puck on level with the barrel as long as possible. Repeat with the First Sleeve for Set 2 Repeat with Both Sleeves for Set 3
Max Out	1	3	Dom	With your normal motion, make 3 swings with just the base bat. Go for your personal bests!

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Counterweight Level 2

Drill	Sets	Reps	Side	Desc
Normal Stance	3	3	Both	Starting with just the base bat, make 3 swings on the dominant side, then 3 swings on the non-dominant side. Repeat with the First Sleeve for Set 2 Repeat with Both Sleeves for Set 3
Top of Zone Swings	3	3	Both	Starting with just the base bat, make swings like you were hitting a ball at the very top of the zone creating as much speed as possible. Repeat with the First Sleeve for Set 2 Repeat with Both Sleeves for Set 3
Long Zone Swings	3	3	Both	With just the base bat, make 3 swings on the dominant side trying to keep the bat as flat as possible during the entire motion. Try having the knob of the bat slightly higher to start and keep the Velo Puck on level with the barrel as long as possible. Repeat with the First Sleeve for Set 2 Repeat with Both Sleeves for Set 3
Max Out	1	3	Dom	With your normal motion, make 3 swings with just the base bat. Go for your personal bests!

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Counterweight Level 3

Drill	Sets	Reps	Side	Desc
Normal Stance	3	3	Both	Starting with just the base bat, make 3 swings on the dominant side, then 3 swings on the non-dominant side. Repeat with the First Sleeve for Set 2 Repeat with Both Sleeves for Set 3
Trail Leg Back	3	3	Both	With just the base bat, put your trail foot back and up on the toe. Make swings trying to create as much bat speed as possible from this position. Repeat with the First Sleeve for Set 2 Repeat with Both Sleeves for Set 3
Top of Zone Swings	3	3	Both	Starting with just the base bat, make swings like you were hitting a ball at the very top of the zone creating as much speed as possible. Repeat with the First Sleeve for Set 2 Repeat with Both Sleeves for Set 3
Max Out	1	3	Dom	With your normal motion, make 3 swings with just the base bat. Go for your personal bests!

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Power Mechanics Protocols

Another pillar of speed production is the mechanics of the player's motion. For these protocols, we are not looking at the "style" of the player's motion. We are looking at the efficient use of Ground Reaction Forces, Sequencing, and Bat Delivery. All of these areas work together to produce the most efficient use of speed.

Notes:

- We always recommend that players go through our dynamic warm-up before training.
- Do each drill sequence 3X from top to bottom in each training session.
- Players will always take a 60-90 second break between each progression of the drill sequence.

Ground Force Level 1 - Optimizing the Kinetic Sequence

Equipment Needed - Force Board/Force Pedal, Velo OverSpeed both Sleeves, Game Bat

Drill	Sets	Reps	Side	Desc
Bat on Chest Loads FP	3	5	Dom	With the bat across your chest, make the transition motion pressing as hard as possible into the FP with your trail heel.
Transition Loads FP	2	3	Dom	With the fully loaded Velo OverSpeed Bat, make small transition motions smashing the FP under the trail heel as hard as possible.
Tee Balls w/Dual FP	1	3	Dom	Set up to a ball on a tee with a FP under the trail heel and under the lead ball. Make fast, quick swings hitting the ball as hard as possible and smashing the trail FP, then the Lead FP.
Tee Balls Short	1	3	Dom	Try to maintain the same feel as the drill above without the force pedals.
Full Swing w/FP	1	3	Dom	Set up to a ball on a tee with a FP under the trail heel and under the lead ball. Make full swings hitting the ball as hard as possible and smashing the trail FP, then the Lead FP.
Full Swings	1	3	Dom	Try to maintain the same feel as the drill above without the force pedals.

Ground Force Level 2 - Optimizing Timing and Point of Application

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Equipment Needed - Force Board/Force Pedal, Velo OverSpeed both Sleeves, Game Bat

Drill	Sets	Reps	Side	Desc
Skaters - explode	1	8	NA	Set up in athletic posture with your feet about shoulder width apart. Jump toward the trail side landing and exploding toward the lead side as far as possible.
Step Back	2	3	Dom	With the fully loaded Velo OverSpeed Bat, make a small counter-rotation forward, then step and plant the back foot, then make your swing. Set 1- Dry Swings, Set 2 from a Tee.
Trail Foot FP Smash	2	3	Dom	Set up with a FP under the trail heel. Make a swing smashing the FP as hard as possible before the bat moves forward. Set 1- Dry Swings, Set 2 from a Tee.
Step-Through	2	3	Dom	With the fully loaded Velo OverSpeed Bat, make 3 swings on the dominant side lifting the lead foot and then stepping through the swing aggressively. Set 1- Dry Swings, Set 2 from a Tee.
Lead Ball FP Smash	2	3	Dom	Set up with a FP under the lead ball. Make full swings hitting the ball as hard as possible and smashing the lead FP during transition. Set 1- Dry Swings, Set 2 from a Tee.
Double Smash	2	3	Dom	Make swings with a FP under the trail heel and lead ball smashing the trail FP first and then the lead FP. Set 1- Dry Swings, Set 2 from a Tee.
Normal Stance	1	3	Dom	From your normal stance, try to maintain the above feels while hitting a ball. This can be from a tee, soft toss, or live pitch.

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Ground Force Level 3 - Magnitude of Force

Equipment Needed - Force Board/Force Pedal, Velo OverSpeed both Sleeves, Game Bat, Medicine Ball (optional)

Drill	Sets	Reps	Side	Desc
Reverse Jumps	1	5	NA	Do a vertical jump with your arms moving up toward the ceiling as you squat and then moving down to the floor fast as you jump. You can optionally do this with a medicine ball.
Reverse Jumps w/rotation	1	5	Dom	Start in an athletic posture. Rotate your upper body and arms back as you squat down and then rotate the upper body and arms forward as you jump. You can optionally do this drill with a medicine ball.
Double Step	2	3	Dom	Starting with your feet together and the fully loaded Velo OverSpeed Bat, allow a small counter-rotation, step back with the trail foot and plant, then let the lead foot come up and plant before the swing. Set 1- Dry Swings, Set 2 from a Tee.
Lead Leg Jump Backs w/FP	1	5	Dom	Start with a FP under the lead ball and the fully loaded Velo OverSpeed Bat. Make a swing pushing so hard in transition that you jump back and away prior to when you would make impact with the ball. Dry swings only.
Rocker Drill w/FP	2	3	Dom	Make swings with a FP under the trail heel and lead ball. Push into the lead FP, then the trail and repeat until you can create a maximum speed swing. Set 1- Dry Swings, Set 2 from a Tee.
Rocker Drill	2	3	Dom	Repeat the above Rocker Drill without the Force Pedals. Set 1- Dry Swings, Set 2 from a Tee.
Normal Stance	1	3	Dom	From your normal stance, try to maintain the above feels while hitting a ball. This can be from a tee, soft toss, or live pitch.

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Sequencing Level 1 - Pelvis/Torso Rotational Speed

Equipment Needed - Velo OverSpeed both Sleeves, Game Bat,

Drill	Sets	Reps	Side	Desc
Hip Twisters	2	5	NA	Hold the bat in front with the end on the ground. Rotate the hips back and forth while keeping the upper body as stable as possible.
Bat on Chest Hip Thrusters	2	5	Dom	With the bat across your chest, get into your starting position and then rotate the hips toward while keeping the upper body back. You can add a small step with the lead foot.
Bat on Chest Swings	1	5	Dom	With the bat across your chest, make swings with a small step focusing on the hips leading the rotation.
Small Step and Go	2	3	Dom	Do a small step with the lead foot and then let the hips rotate through the swing. Set 1- Dry Swings, Set 2- from a tee.
Big Step and Go	2	3	Dom	Make a much bigger and more powerful step and then let the hips rotate aggressively through the swing. Set 1- Dry Swings, Set 2 from a Tee.
Normal Stance	1	3	Dom	From your normal stance, try to maintain the above feels while hitting a ball. This can be from a tee, soft toss, or live pitch.

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Sequencing Level 2 - Torso/Lead upper arm connection.

Equipment Needed - Velo OverSpeed both Sleeves, Game Bat, Towel

Drill	Sets	Reps	Side	Desc
Lead arm towel thrusts	2	5	Dom	Hold a towel between the lead upper arm and the chest. From your starting position with the fully loaded Velo OverSpeed Bat, make initial transitions smashing the towel.
Lead arm towel	2	3	Dom	From the same position as above with your game bat, make swings smashing the towel during transition. Set 1- Dry Swings, Set 2 from a Tee.
Chest pump and go	2	3	Dom	From your starting position with your game bat, make 2-3 pump motions feeling the pressure between your chest and lead upper arm and then swing. Set 1- Dry Swings, Set 2 from a Tee.
Big Step and Go	2	3	Dom	Make a much bigger and more powerful step and then let the hips rotate aggressively through the swing. Set 1- Dry Swings, Set 2 from a Tee.
Normal Stance	1	3	Dom	From your normal stance, try to maintain the above feels while hitting a ball. This can be from a tee, soft toss, or live pitch.

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Bat Delivery Level 1

Equipment Needed: Velo OverSpeed Bat w/both sleeves, Game Bat, Velo Puck

Drill	Sets	Reps	Side	Desc
Lead Arm Only, Slow delivery swings	1	5	Dom	With the lead arm only, make a swing letting the knob of the bat lead through the zone. The lead arm should rotate toward the palm facing up through the zone.
Trail Arm Only, Slow delivery swings	1	5	Dom	With the trail arm only, make swings letting the trail arm extend from the elbow through the zone.
Pump, Handle Lead w/Puck	2	3	Dom	With the Velo Puck on your game bat, Make 2-3 pumps and then swing making sure the knob leads in transition. Set 1- Dry Swings, Set 2 from a Tee.
Velo Puck Swings, high tee	2	3	Dom	With the Velo Puck on your game bat, make swings at a high tee. Make sure the handle and barrel stay as level as possible throughout this motion. Set 1- Dry Swings, Set 2 from a Tee.
Velo Puck Swings, tee Forward	2	3	Dom	With the Velo Puck on your game bat, make swings at a forward tee. Make sure the handle and barrel stay as level as possible throughout this motion. Set 1- Dry Swings, Set 2 from a Tee.
Counter-Rotation Swings	2	3	Dom	With your game bat, make a counter-rotation toward the ball and then a full swing. Set 1- Dry Swings, Set 2 from a Tee.
Sprint	2	5	Dom	With your game bat, make 5 swings in as fast of succession as possible.
Normal Stance	1	3	Dom	From your normal stance, try to maintain the above feels while hitting a ball. This can be from a tee, soft toss, or live pitch.

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Exit Velo Application Drills

The end goal of increasing bat speed is seeing more exit velo in game. These protocols are designed to create optimal transfer of your bat speed training to live balls.

Exit Velo Application Level 1 - Tee only

Equipment Needed: Velo Overspeed Bat w/sleeves, Game Bat

Drill	Sets	Reps	Side	Desc
Fully Loaded Velo Swings	1	3	Dom	Make 3 aggressive swings with the fully loaded Velo Bat.
Tee Hits	1	2	Dom	Hit two balls from a tee.
1 Sleeve Velo Swings	1	3	Dom	Make 3 aggressive swings with the Velo Bat Medium Sleeve
Tee Hits	1	2	Dom	Hit two balls from a tee.
Base Velo Swings	1	3	dom	Make 3 aggressive swings with the Velo Bat base only.
BREAK				2 Minute Break
3 Strike Ramp Up	VAR	VAR	Dom	Hit a ball from a tee and note your bat speed and Exit Velo (if possible). Hit a second ball. If either your bat speed or Exit Velo increases, move on to a third ball. If they both go down, you have one strike. Hit another ball and repeat this process until you have 3 strikes.

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Exit Velo Application Level 2 - Tee, Soft Toss, or Live Pitch

Equipment Needed: Velo Overspeed Bat w/sleeves, Game Bat

Drill	Sets	Reps	Side	Desc
Fully Loaded Velo Swings - Step and Go	1	3	Dom	Make 3 aggressive swings with the fully loaded Velo Bat.
Hits	1	2	Dom	Hit two balls from a tee, soft toss, or live pitch
1 Sleeve Velo Swings - Step and Go	1	3	Dom	Make 3 aggressive swings with the Velo Bat Medium Sleeve
Hits	1	2	Dom	Hit two balls from a tee, soft toss, or live pitch
Base Velo Swings - Step and Go	1	3	dom	Make 3 aggressive swings with the Velo Bat base only.
BREAK				2 Minute Break
3 Strike Ramp Up	VAR	VAR	Dom	Hit a ball from a tee, soft toss, or live pitch and note your bat speed and Exit Velo (if possible). Hit a second ball. If either your bat speed or Exit Velo increases, move on to a third ball. If they both go down, you have one strike. Hit another ball and repeat this process until you have 3 strikes.

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Exit Velo Application Level 3 - Tee, Soft Toss, or Live Pitch

Equipment Needed: Velo Overspeed Bat w/sleeves, Game Bat, Velo Puck

Drill	Sets	Reps	Side	Desc
Fully Loaded Velo Swings	1	3	Dom	Make 3 aggressive swings with the fully loaded Velo Bat.
Hits	1	2	Dom	Hit two balls from a tee, soft toss, or live pitch
Velo Bat 1 Sleeve + Velo Puck	1	3	Dom	Make 3 aggressive swings with the fully loaded Velo Bat.
BREAK				2 Minute Break
3 Strike Ramp Up	VAR	VAR	Dom	Hit a ball from a tee, soft toss, or live pitch and note your bat speed and Exit Velo (if possible). Hit a second ball. If either your bat speed or Exit Velo increases, move on to a third ball. If they both go down, you have one strike. Hit another ball and repeat this process until you have 3 strikes.

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Warm-Up Protocols

Dynamic Warm-Up - This one will fall into the fitness category. This will be a full body dynamic warm-up that will be great before training or as part of the pre-game warm-up process.

Pre-Game Warm-Up

Equipment Needed - Velo OverSpeed Bat w/sleeves

Drill	Sets	Reps	Side	Desc
Fully Loaded Velo Bat Gradual Increase	2	10	Both	Starting on the non-dom side for set 1, make 10 swings starting at about 50% of max and ending close to 100%. Repeat this process on the dominant side for set 2.
Velo 1 Sleeve Swings	2	3	Both	Starting on the non-dom side for set 1, make 3 aggressive swings. Repeat on the dominant side.
Velo Base Swings	2	3	Both	Starting on the non-dom side for set 1, make 3 aggressive swings. Repeat on the dominant side.
BREAK				60-90 Seconds
Game Bat Swings	1	3	Dom	Make 3 Swings with your game bat on the dominant side only.

On Deck Protocol

Equipment Needed: Velo OverSpeed Bat w/sleeves, Game Bat

Drill	Sets	Reps	Side	Desc
Velo 1 Sleeve Swings	2	3	Both	Starting on the non-dom side for set 1, make 3 aggressive swings. Repeat on the dominant side.
Game Bat Swings	1	3	Dom	Make 3 Swings on the Dominant Side with your game bat.
REPEAT DRILL 1/2				Repeat these two drills as needed

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Supporting Physical Fitness

Equipment Pack Needed: FMT Bands, Rip Trainer Type, Suspension Trainer, Balance Board, Medicine Ball, Kettle Bell.

Lower Body Mobility (Ankles, hips)
Core Mobility (T-Spine rotation, Side Bend, Flexion/Extension)
Upper Body Mobility (Neck, Shoulders, Wrists)
Balance/Stability Level 1
Balance/Stability Level 2
Balance/Stability Level 3
Lower Body Strength Level 1
Lower Body Strength Level 2
Core Strength Level 1
Core Strength Level 2
Upper Body Strength Level 1
Upper Body Strength Level 2

Lower Body Jump Power
Core Pull Power
Upper Body Push Power

ASAP - Pre-Training Advanced Speed Activation